

RECENT PROGRESSES OF ACCELERATED MRI USING ANNIHILATING FILTER-BASED LOW-RANK INTERPOLATION

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ABSTRACT

Recently, an annihilating filter based low-rank Hankel matrix approach (ALOHA) was proposed as a general framework for sparsity-driven k-space interpolation method for compressed sensing MRI (CS-MRI). The principle of ALOHA framework is based on the fundamental duality between the transform domain sparsity in the primary space and the low-rankness of *weighted* Hankel matrix in Fourier domain, which converts CS-MRI to a k-space interpolation problem using structured matrix completion. In this review, we explain the theory behind ALOHA. Experimental results with in vivo data for multi-coil dynamic imaging, parametric mapping as well as Nyquist ghost correction confirmed that the proposed method has potential to be a general solution of various MR imaging problems.

Index Terms— ALOHA, annihilating filter, structured low rank matrix completion, parallel MRI, T1/T2 parametric mapping, Nyquist ghost correction

1. INTRODUCTION

Magnetic resonance imaging (MRI) is an imaging system that sequentially acquires k-space data corresponding to the Fourier transform of an object. This enables us to apply various advanced signal processing techniques. Recently, compressed sensing theory [1, 2] has been used in accelerated MRI [3, 4, 5, 6, 7]. Compressed sensing algorithms can recover the signals from subsampled k-space data by using the sparsity of an unknown image in total variation (TV) or wavelet transform domains with incoherent sampling schemes such as Gaussian random or Poisson disc.

On the other hand, parallel MRI (pMRI) [8, 9] explores the diversity in the coil sensitivity maps that are multiplied by an unknown image. Those give additional spatial information for the unknown image, resulting in accelerated MR data acquisition through k-space sample reduction. Parallel imaging algorithms such as SENSE (sensitivity encoding) [8] or GRAPPA (generalized autocalibrating partially parallel acquisitions) [9] require regularly sampled k-space data

for computationally efficient reconstruction. Moreover, additional k-space data, the so-called auto calibration (ACS) lines, are often required to estimate the coil sensitive maps or GRAPPA kernels [9].

In order to overcome these difficulties, calibration-less parallel imaging techniques have been extensively explored, among which SAKE (simultaneous autocalibrating and k-space estimation)[10] and LORAKS(Low-rank modeling of local k-space neighborhoods)[11] were proposed. In SAKE, the missing k-space elements are reconstructed by imposing the data consistency and the structural maintenance constraints of the block Hankel structure matrix. However, the origin of the low rankness in the Hankel structured matrix for the case of a single coil measurement was not extensively investigated. LORAKS was proposed to reconstruct missing elements by using low-rankness of augmented Hankel matrix with finite support or slow-varying phase conditions. However, it is not clear how the existing theory can deal with large classes of image models that are not finite supported but can be sparsified using various transforms such as wavelet transforms or total variations (TV), etc.

Here, we review our recent theory that unifies and generalizes k-space low-rank methods to also allow for transform sparsity models which are critical for practical MRI applications. In this paper, we first show that the transform domain sparsity in the signal space can be directly related to the existence of *annihilating filters* [12] in the *weighted* k-space. Interestingly, the commutative relation between an annihilating filter and weighted k-space measurements provides a rank-deficient Hankel structured matrix, whose rank is determined by the sparsity level of the underlying signal in a transform domain. Therefore, by performing a low-rank matrix completion approach, the missing weighted k-space data in the Hankel structured matrix can be recovered, after which the original k-space data can be recovered by removing the weights. With several real world MRI applications, we demonstrate the generality of the proposed approach in the context of MR imaging.

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2. FUNDAMENTAL DUALITY

Consider a signal $f(x)$ that is composed of a set of k -Diracs. Then, it is well-known that the corresponding Fourier discrete spectrum $\hat{f}[k]$ can be cancelled out by a $k+1$ length annihilating filter $\hat{h}[k]$ such that the corresponding discrete counterpart is given by

$$(\hat{h} * \hat{f})[n] = \sum_{l=0}^k \hat{h}[l] \hat{f}[n-l] = 0, \quad n \in \mathbb{Z}, \quad (1)$$

in [12].

Accordingly, by defining n_1 such that $\min(n_1 - \kappa + 1, \kappa) \geq k$, we can construct the following matrix equation:

$$\mathcal{H}(\hat{f}) \bar{\mathbf{h}} = \mathbf{0}, \quad (2)$$

where the Hankel structure matrix $\mathcal{H}(\hat{\mathbf{f}}) \in \mathbb{C}^{(n_1 - \kappa + 1) \times \kappa}$ is constructed as

$$\mathcal{H}(\hat{f}) = \begin{bmatrix} \hat{f}[0] & \hat{f}[1] & \cdots & \hat{f}[\kappa-1] \\ \hat{f}[1] & \hat{f}[2] & \cdots & \hat{f}[\kappa] \\ \vdots & \vdots & \ddots & \vdots \\ \hat{f}[n_1 - \kappa] & \hat{f}[n_1 - \kappa + 1] & \cdots & \hat{f}[n_1 - 1] \end{bmatrix} \quad (3)$$

One of the most important results from [13] is then, for $\kappa \geq k + 1$, we can further show the following key result [13]:

$$\text{RANK} \mathcal{H}(\hat{f}) = k. \quad (4)$$

which implies that as long as the annihilating filter size is bigger than the sparsity level, the rank of the associated Hankel matrix is always equal to the sparsity level. Hence, if we make κ sufficiently large, the corresponding Hankel matrix $\text{RANK} \mathcal{H}(\hat{f})$ becomes low-ranked.

In general, signals may not be sparse in the image domain, but can be sparsified in a transform domain [14]. For example, a piecewise constant signal f can be sparsified using the first order differentiation:

$$g(x) = Df(x) = \sum_{j=0}^{k-1} c_j \delta(x - x_j) \quad (5)$$

where D denotes the first order differential operator. The signals with the total variation belong to this class of signals. In this case, based on our discussion so far, we know that there exists annihilating filter relationship for $\hat{g}[k]$, and the associated Hankel matrix with respect to its discretized signal $\hat{g}[k]$ is rank-deficient. Thanks to the k -space measurements in MRI, we can convert $\hat{f}(\omega)$ into $\hat{g}(\omega)$ by simple multiplications given as

$$\hat{g}(\omega) = (j\omega) \hat{f}(\omega), \quad (6)$$

and the weighting $(j\omega)$ is deterministic, so we can construct a Hankel structured matrix from the weighted k -space measurements, and the resulting Hankel matrix is low-ranked.

Therefore, if the missing elements of $\hat{g}[n]$ is estimated via a low rank Hankel matrix completion, the missing k -space data $\hat{f}[n]$ can be obtained by dividing them with the deterministic weighting. Therefore, we are interested in solving the following weighted low-rank matrix completion problem:

$$(P) \quad \min \quad \|\mathcal{H}(M)\|_* \quad (7)$$

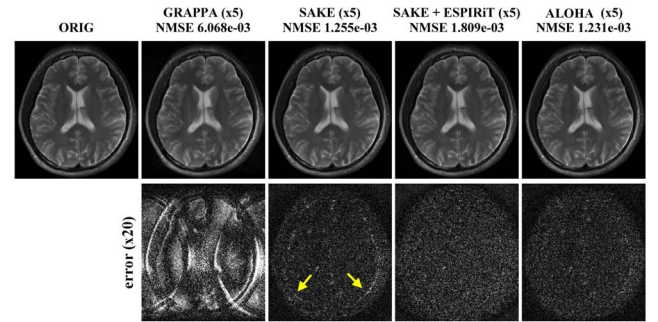
subject to $P_\Omega(M) = P_\Omega(L \odot S)$

where M is target to be reconstructed, P_Ω is the sampling operator where real data is acquired, L is sparsifying weighting, $\odot$ is Hardamard product, and S is the measurement.

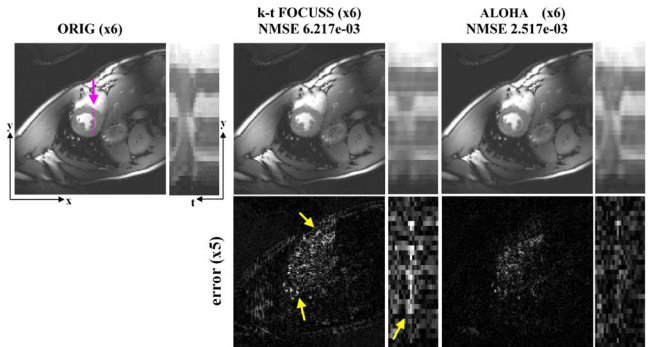
3. APPLICATIONS TO CS-MRI

In this section, for an easy explanation, we revisit several MR applications of the theory.

3.1. Accelerated MRI



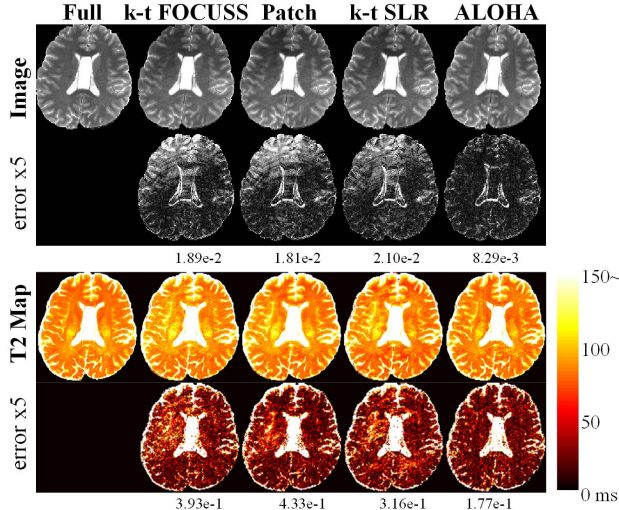
(a) parallel imaging (4coils)



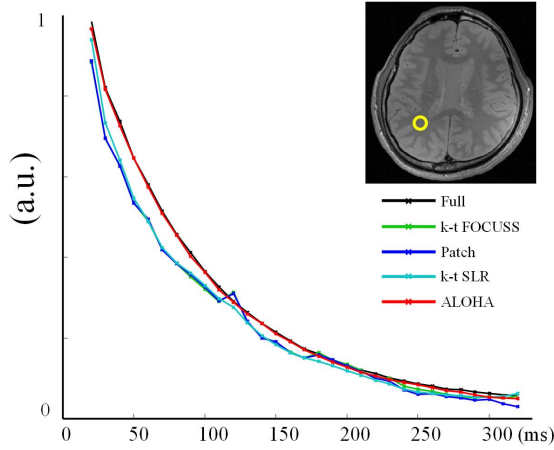
(b) parallel dynamic imaging (4coils)

Fig. 1: Retrospective reconstruction results from (a) 5 fold accelerated pMRI ($x-y$ sparsity) (b) 6 fold accelerated pMRI ($x-f$ sparsity). Purple line in (b) denotes the regions corresponding to $y-t$ cross sections that are magnified along temporal axis. The second rows show the residual images.

Recall that ALOHA establishes the duality between rank of weighted Hankel matrix in primary domain and sparsity in reciprocal domain. This property is easily extended into 2D



(a) Reconstruction - x8 accelerated real in vivo



(b) Signal decay curves of T2 images (x8)

Fig. 2: (a) Reconstructed ME-SE images (Echo=7th) acquired by 8 times accelerated *in vivo* ME-SE sequences using k-t FOCUSS [5], Patch (patch-based low-rank approach)[6], k-t SLR [15], and ALOHA, respectively. (b) Signal decaying curves from various algorithms are plotted.

domain. Additionally, in $x - f$ domain of dynamic imaging, the sparsified spatio-temporal profiles can be easily obtained in the similar manner. Therefore, we applied ALOHA algorithm for (P) (Eq. (7)) in $k_x - k_y$ domain and $k - t$ domain for accelerated MR imaging. As shown in Fig. 1, the performance of ALOHA outperformed the other compressed sensing based algorithms in Gaussian sampling distribution.

3.2. Accelerated T1/T2 parametric mapping

To apply ALOHA for parameter mapping, the property of a dynamic parameter image data was further investigated [16].

Here, y denote the read-out direction and $\nu(x, t)$ is the spatio-temporal volume of unknown MR signals at a given y . In accelerated MR data acquisition, the frequency encoding along k_y is done with full samples, and we use an undersampling along the phase encoding direction in the (k, t) dimension. Therefore, we are only interested in spatio temporal modeling of $\nu(x, t)$ at a fixed y . Here, assume that the temporal Fourier transform can sparsify the signal, and the spatio temporal MR signal can be modelled as $x - f$ sparsity ALOHA model. Again, in Fig. 2, we confirmed that ALOHA model could be easily extended into accelerated MR parametric mapping applications.

3.3. Nyquist ghost correction

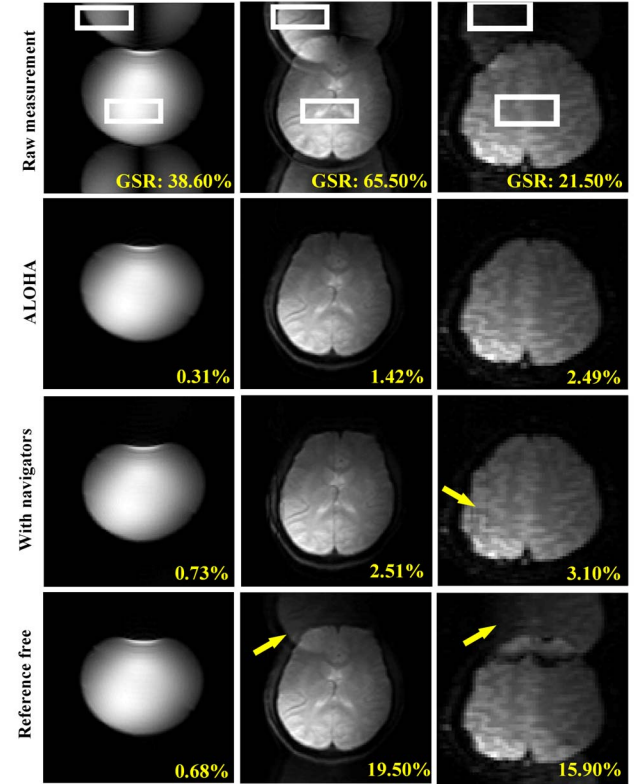


Fig. 3: Ghost correction results from three different methods: ALOHA without navigators, correction with navigators, correction without navigators [17]. The white boxed areas correspond to calculation region for the ghost-to-signal ratio (GSR) values. The inlayed numbers indicate the GSR values.

Because Nyquist ghost correction occurred by eddy current from abrupt change of gradient field [18, 19], this effect is unavoidable in general EPI sequence [20]. However, we simply changed Nyquist ghost correction problem into sub-

sampled Fourier measurements with two measurements including phase differences by decoupling even and odd lines data [20]. Those decoupled data has jointly shared support between even and odd lines images.

To calculate the Hankel matrix from $S_{n,\Delta} = S_{n,+} - S_{n,-}$, here we provide an equivalent formulation that can be easily exploited. More specifically, the low rankness implies that there exists an annihilating filter vector $\mathbf{h}$ such that

$$\mathcal{H}(S_{n,\Delta})\mathbf{h} = (\mathcal{H}(S_{n,+}) - \mathcal{H}(S_{n,-}))\mathbf{h} = \mathbf{0} \quad (8)$$

This implies that

$$\mathcal{Y} \begin{bmatrix} \mathbf{h} \\ -\mathbf{h} \end{bmatrix} = \mathbf{0},$$

where

$$\mathcal{Y} = \begin{bmatrix} \mathcal{H}(S_{n,+}) & \mathcal{H}(S_{n,-}) \end{bmatrix}. \quad (9)$$

Accordingly, the concatenated matrix $\mathcal{Y}$, which is obtained by stacking the two Hankel matrices from even and odd echoes, is low-ranked. Therefore, using a low rank structured matrix completion algorithm for $\mathcal{Y}$, we can fill in the missing odd and even k-space lines for $S_{n,+}$ and $S_{n,-}$. Then, the final ghost-free image can be obtained as a sum of squares (SSoS) of the even and odd line image reconstructions (see Fig. 3).

ALOHA-based ghost artifact correction for EPI data can be formulated by the following nuclear norm minimization problem:

$$\begin{aligned} (P') \quad & \min \quad \|\mathcal{H}(M_+) \mathcal{H}(M_-)\|_* \\ \text{subject to} \quad & P_{\Omega_+}(M_+) = P_{\Omega_+}(S_{n,+}) \\ & P_{\Omega_-}(M_-) = P_{\Omega_-}(S_{n,-}) \end{aligned}$$

where $\|\cdot\|_*$ denotes a nuclear norm, and P_{Ω} is a projection operator on the sampling index set Ω , and Ω_- and Ω_+ denote the sampling indice for odd and even index, respectively. As shown in Fig. 3, the reconstructed image from ALOH-based ghost correction showed the best GSR values compared with other methods in several objects.

4. CONCLUSION

In this paper, we reviewed our recent framework for annihilating filter based low-rank Hankel matrix approach (ALOHA) for several MR imaging applications. In particular, various MR applications such as accelerated MR imaging in dynamic and static objects, accelerated parametric mapping, and even Nyquist ghost correction, were successfully solved using ALOHA. We believe that ALOHA framework can provide good opportunities to overcome fundamental limitations related with accelerated MR acquisition and inherent artifacts.

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